THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the column algorithm for addition, a carry is generated when a column sum is: 1

(a) greater than 9 **(b)** less than 9 **(c)** exactly 0 **(d)** an even number

2. Set total = 0. For i = 1, 2, …, 8: if i is even, add i to total. The final value of total is: 1

(a) 22 **(b)** 21 **(c)** 20 **(d)** 19

3. Which of these must be true of every step of an algorithm? 1

(a) it depends on luck **(b)** it can be skipped

(c) it is clear and unambiguous **(d)** it is different each time

4. Adding 10 and 49 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 1 **(b)** 0 **(c)** 2 **(d)** 3

5. Using the column addition algorithm, $341 + 379$ equals: 1

(a) 720 **(b)** 820 **(c)** 710 **(d)** 730

6. The divisor algorithm applied to 17 produces a list with how many entries? 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

7. The value of $43 \bmod 9$ (remainder on dividing 43 by 9) is: 1

(a) 8 **(b)** 9 **(c)** 7 **(d)** 6

8. The greatest of the common divisors of 22 and 36, found from their divisor lists, is: 1

(a) 4 **(b)** 3 **(c)** 1 **(d)** 2

9. The remainder when 29 is divided by 6 is written: 1

(a) $29 \bmod 6 = 6$ **(b)** $29 \bmod 6 = 5$ **(c)** $29 \bmod 6 = 0$ **(d)** $29 \bmod 6 = 4$

10. Set total = 0. For i = 1, 2, …, 20: if i is even, add i to total. The final value of total is: 1

(a) 111 **(b)** 110 **(c)** 112 **(d)** 109

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find $73 \bmod 5$ and $78 \bmod 5$. What do you notice? 2

12. Add 446 and 397 using the column algorithm. State in which columns a carry is produced. 2

13. Run the divisor algorithm for 20: check j = 1, 2, …, 20. Write the list of divisors and say how many numbers are checked. 2

14. Find $66 \bmod 6$ and $72 \bmod 6$. What do you notice? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find gcd(38, 10) by listing the divisors of both numbers. 3

16. Apply Euclid's subtraction algorithm to find gcd(32, 24). Write the sequence of pairs and count the subtractions. 3

17. Add 9110 and 2211 using the column algorithm. State in which columns a carry is produced. 3

OR

Find gcd(25, 40) by listing the divisors of both numbers. 3

18. Apply Euclid's subtraction algorithm to find gcd(52, 40). Write the sequence of pairs and count the subtractions. 3

OR

Add 3551 and 3723 using the column algorithm. State in which columns a carry is produced. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace this procedure: total = 0; for i = 1 to 8, if i is even add i to total. Write the values added and the final total. 5

OR

Find gcd(66, 40) by listing the divisors of both numbers. 5

20. Trace this procedure: total = 0; for i = 1 to 13, if i is even add i to total. Write the values added and the final total. 5

OR

Find gcd(50, 85) by listing the divisors of both numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND